Segurança de Redes

Network Security

Mestrado em Cibersegurança (MCS)

Mestrado em Redes de Computadores (MRC)

Departamento de Informática, Universidade do Minho

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1° Semestre (26/27)

Adapted from Prof. Paulo Carvalho's (Dept. Informática – UMinho) slides, which were originally based on:
J.F Kurose and K.W. Ross, "Computer Networking: A Top-Down Approach, 8th ed., Pearson, 2020. Teaching material.

Routing Security



Image created with ChatGPT (OpenAI GPT-6.1)

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Outline

- Routing Security
  - Interdomain Routing
    - BGP Concepts (review)
    - BGP Security Vulnerabilities
    - BGP Security Support Technologies
  - Intradomain Routing
    - IGP Concepts (review)
    - IGP Vulnerabilities and Attacks
    - Example of OSPF LSA Injection Attack
    - Summary



Why Does Routing Security Matter?

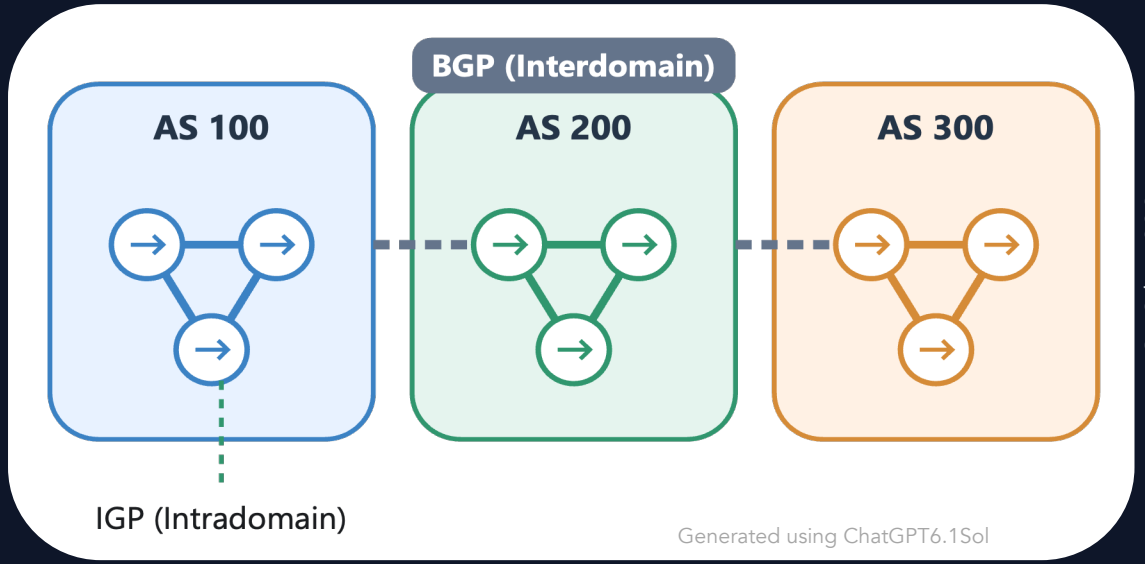
Routing protocols determine how packets travel across networks

- Routers exchange routing information to discover reachable networks and select paths.
- Incorrect or malicious routing information can lead to:
  - Traffic redirection or interception
  - Network outages and unreachable destinations (blackholing)
  - Routing instability and performance degradation
- Routing security aims to ensure the authenticity and integrity of routing information.

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Internet Routing: Autonomous Systems

- The Internet consists of interconnected networks managed by different organizations.
- These networks are organized into **Autonomous Systems (ASes)**:
  - Networks under a common administrative routing policy (e.g., ISPs, universities, enterprises)
  - Each identified by an Autonomous System Number (ASN)
- Two main routing domains:
  - **Interdomain routing**: routing between ASes (BGP)
  - **Intradomain routing**: routing within an AS (e.g., OSPF, IS-IS)



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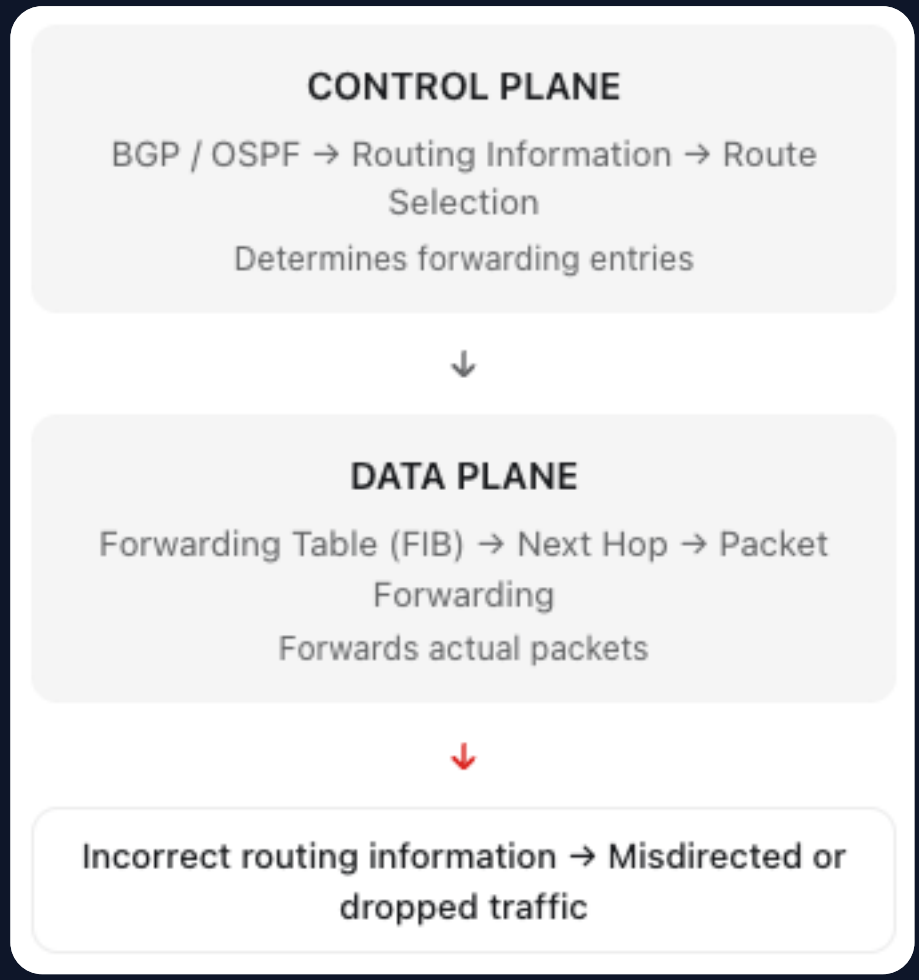
Control Plane vs. Data Plane

Control Plane — Where should packets go?

- Determines network routes and maintains routing information.
- Routing protocols (e.g., BGP, OSPF) exchange information to establish routes.
- Routing decisions determine which paths are installed for forwarding.

Data Plane — How are packets forwarded?

- Forwards packets based on the router's forwarding table (FIB).
- Uses destination IP addresses and forwarding entries to select the next hop.
- Operates on actual network traffic.

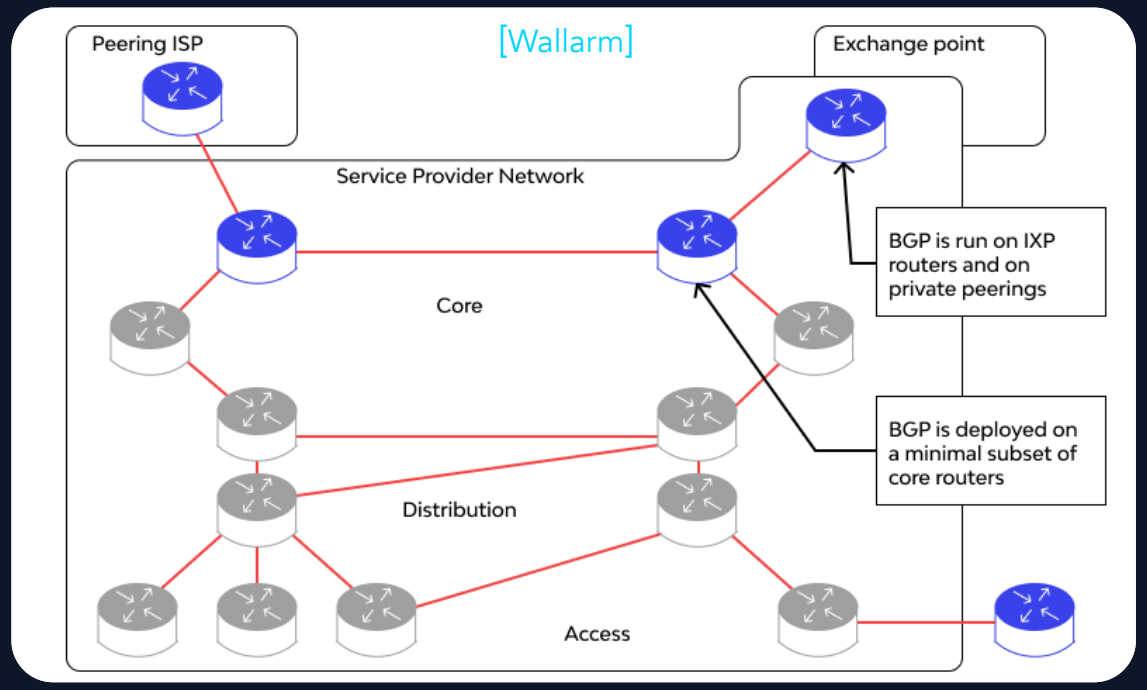


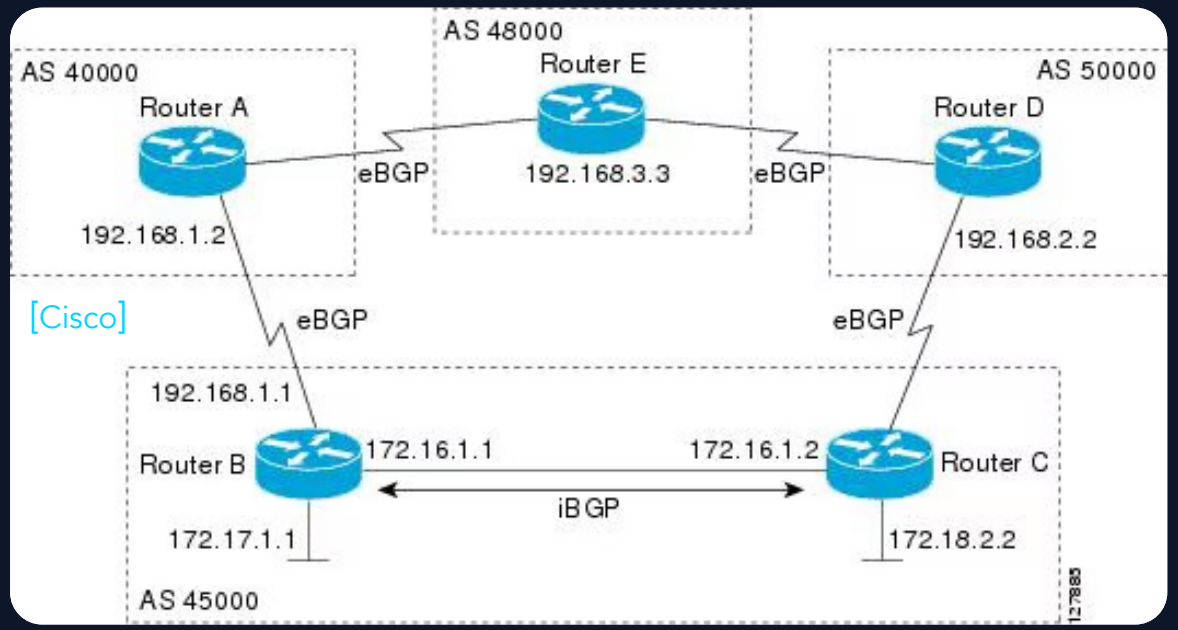
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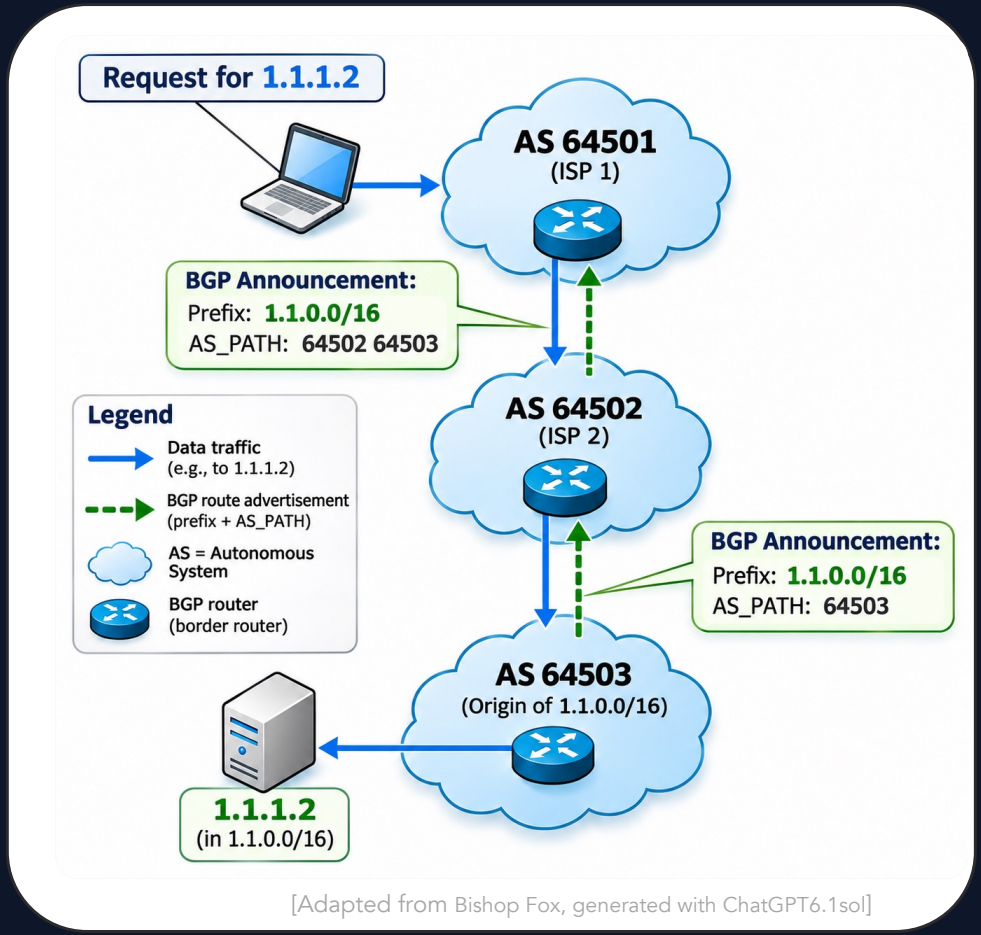
BGP Concepts (review)

- BGP (Border Gateway Protocol) and its role on the Internet [RFC 4271]
  - primary interdomain routing protocol
  - enables different Autonomous Systems (ASes) to exchange routing information and determine the best paths for delivering packets globally
- each AS is identified by a unique number (ASN) and represents a network managed by a single administrative entity (e.g., ISP, enterprise, university)
- BGP operates as a path-vector protocol, exchanging information about reachable networks (IP prefixes) and the AS path required to reach them
- Peering Relationships:
  - external BGP (eBGP) – connects ASes across administrative boundaries
  - internal BGP (iBGP) – distributes routing information within an AS
- BGP selects paths based on policies and metrics such as AS path length

BGP Concepts







[Adapted from Bishop Fox, generated with ChatGPT6.1sol]

Examples of BGP Deployment

BGP Security Concerns

- Originally, BGP was designed with minimal security considerations
  - Trust-based model
    - assumes that all peers behave correctly and do not propagate incorrect routing info
  - Lack of Routing Information Authentication
    - BGP does not inherently provide cryptographic authentication of route announcements, allowing false routing information to be accepted (spoofing).
  - Global Propagation of Faults
    - errors or malicious changes in one AS can propagate globally, affecting connectivity
  - No Built-In Validation of Route Announcements
    - does not verify the ownership of IP prefixes being advertised, enabling attackers to hijack prefixes or leak routes
  - Susceptibility to Session Attacks
    - BGP sessions between routers are vulnerable to interception and manipulation (e.g., TCP session hijacking or reset attacks)

BGP Security Vulnerabilities

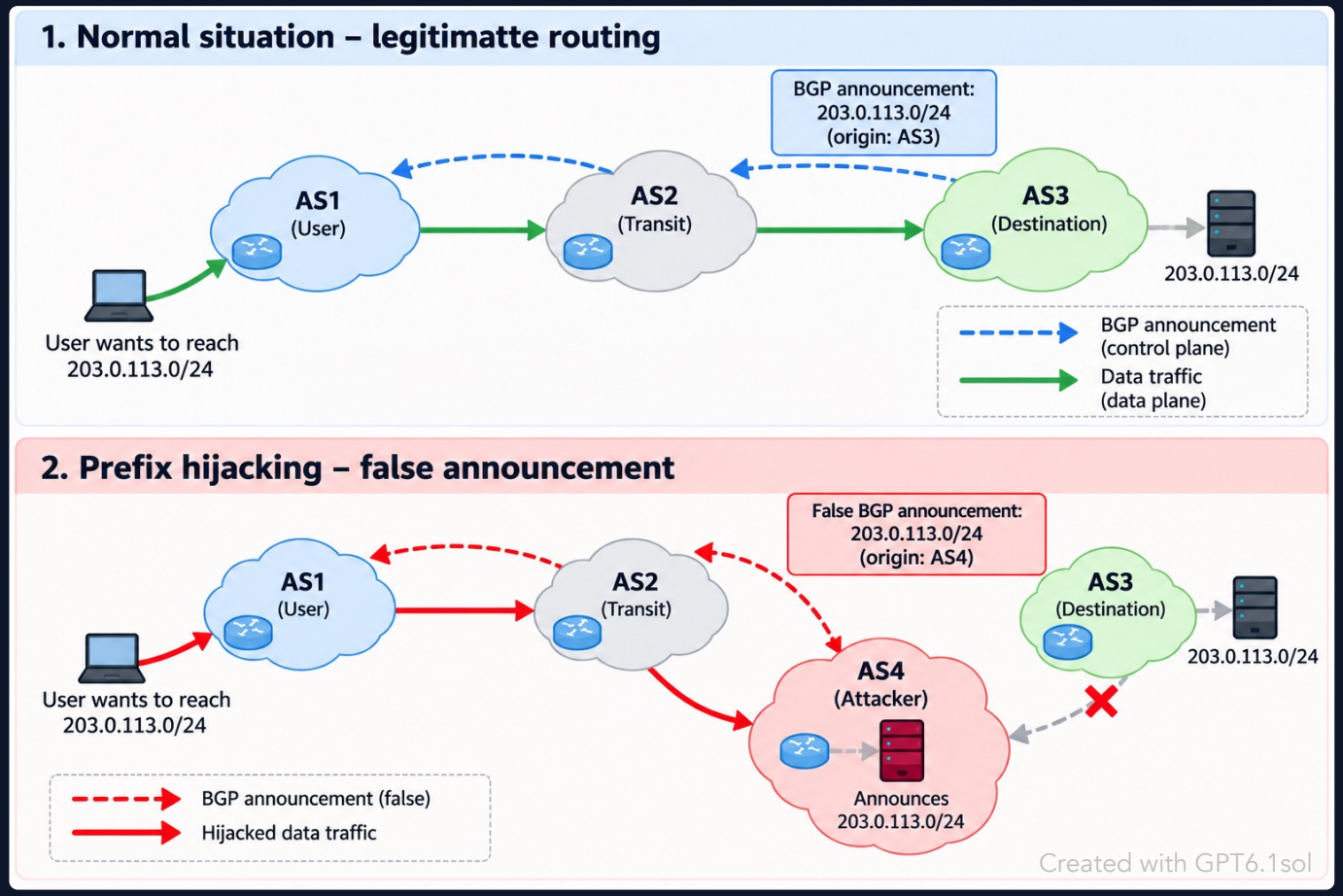
1. Prefix Hijacking

- attackers advertise IP prefixes they do not own, redirecting traffic meant for legitimate networks to their own AS
- may lead to (i) interception or modification of traffic; (ii) traffic blackholing (making the destination unreachable)

Mitigation

- Resource Public Key Infrastructure (RPKI) – validates the origin AS of a route by associating IP prefixes with ASes; uses cryptographic signatures to verify route announcements
- Route Filtering – ISPs can configure prefix filters to ensure only valid routes are propagated
- BGP Monitoring Tools – use tools to detect anomalies in prefix advertisements (e.g., BGPmon or ARTEMIS)

BGP Security Vulnerabilities
Example of BGP Prefix Hijacking

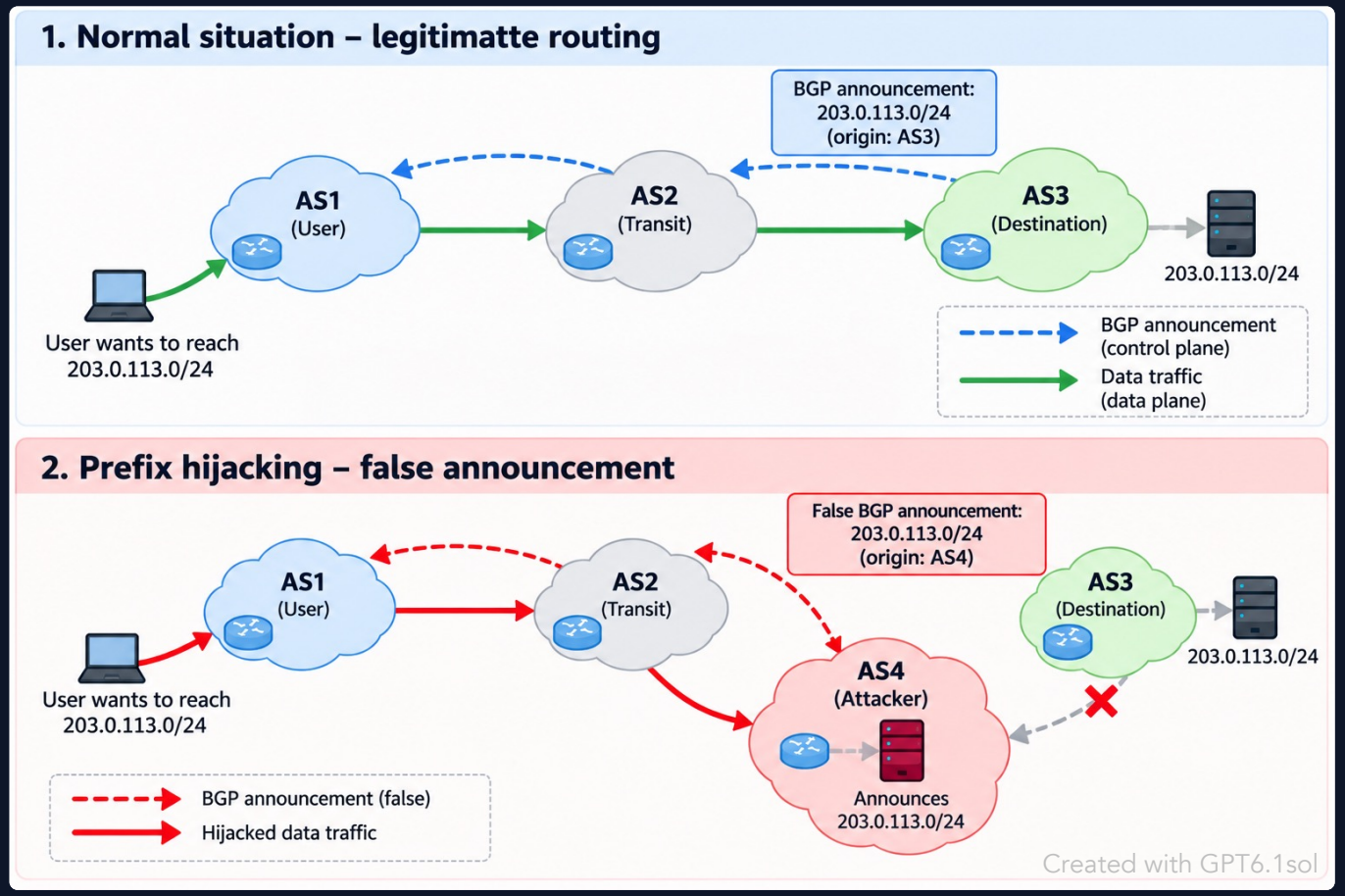


Normal situation (top diagram)

- AS3 legitimately owns and announces the prefix 203.0.113.0/24.
- The BGP announcement propagates from AS3 → AS2 → AS1 (blue dashed arrows).
- AS1 learns that the destination network is reachable through AS2 and AS3.
- When a user sends packets to this network, traffic follows AS1 → AS2 → AS3 (green arrows).
- Notice the distinction: BGP announcements belong to the control plane, while actual packet forwarding belongs to the data plane.

BGP Security Vulnerabilities

Example of BGP Prefix Hijacking



Prefix hijacking (bottom diagram)

- An attacker controlling AS4 falsely announces the same prefix, 203.0.113.0/24, claiming to originate it.
- This false announcement propagates through AS2 to AS1 (red dashed arrows).
- AS2 may accept the false route and prefer it over the legitimate route, depending on its BGP routing policies.
- As a result, AS1 forwards packets through AS2, but instead of reaching AS3, traffic is redirected to AS4 (solid red arrows).
- The attacker now receives traffic intended for the legitimate server.

BGP Security Vulnerabilities

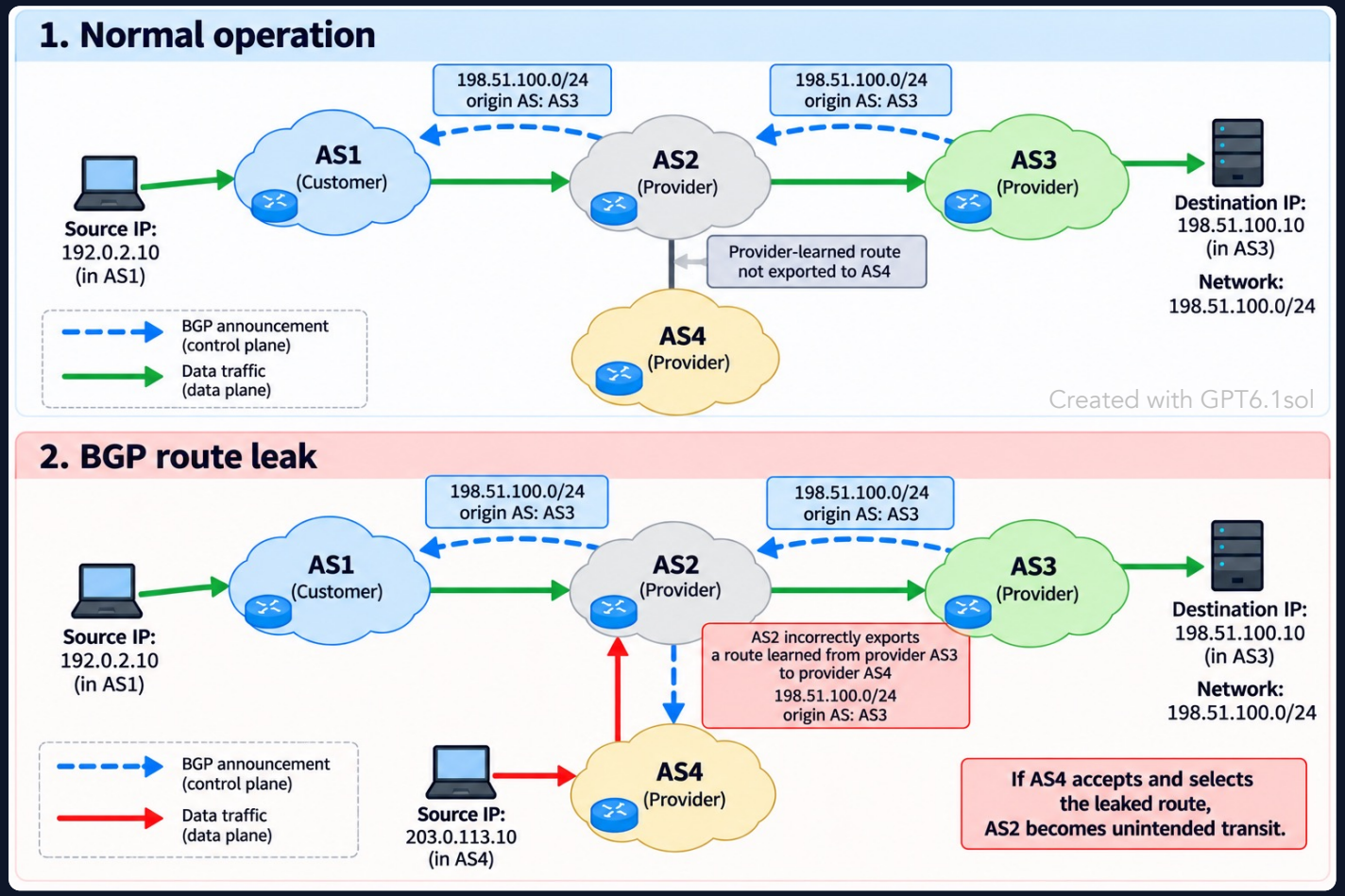
2. Route Leaks

- misconfigured or malicious ASes improperly propagate routes intended for a limited scope (e.g., private peering) to the global BGP tables
- leads to suboptimal routing, i.e., traffic taking inefficient paths
- overloading of intermediate ASes, causing performance degradation

Mitigation

- Route Policy Enforcement – configure strict export policies to control which routes are shared with peers (use the NO_EXPORT community attribute to prevent propagation)
- MANRS (Mutually Agreed Norms for Routing Security)
  - set of guidelines and tools for prevention of route leaks (https://manrs.org)
- BGPsec – provides cryptographic validation of the AS path to ensure routes follow authorized paths

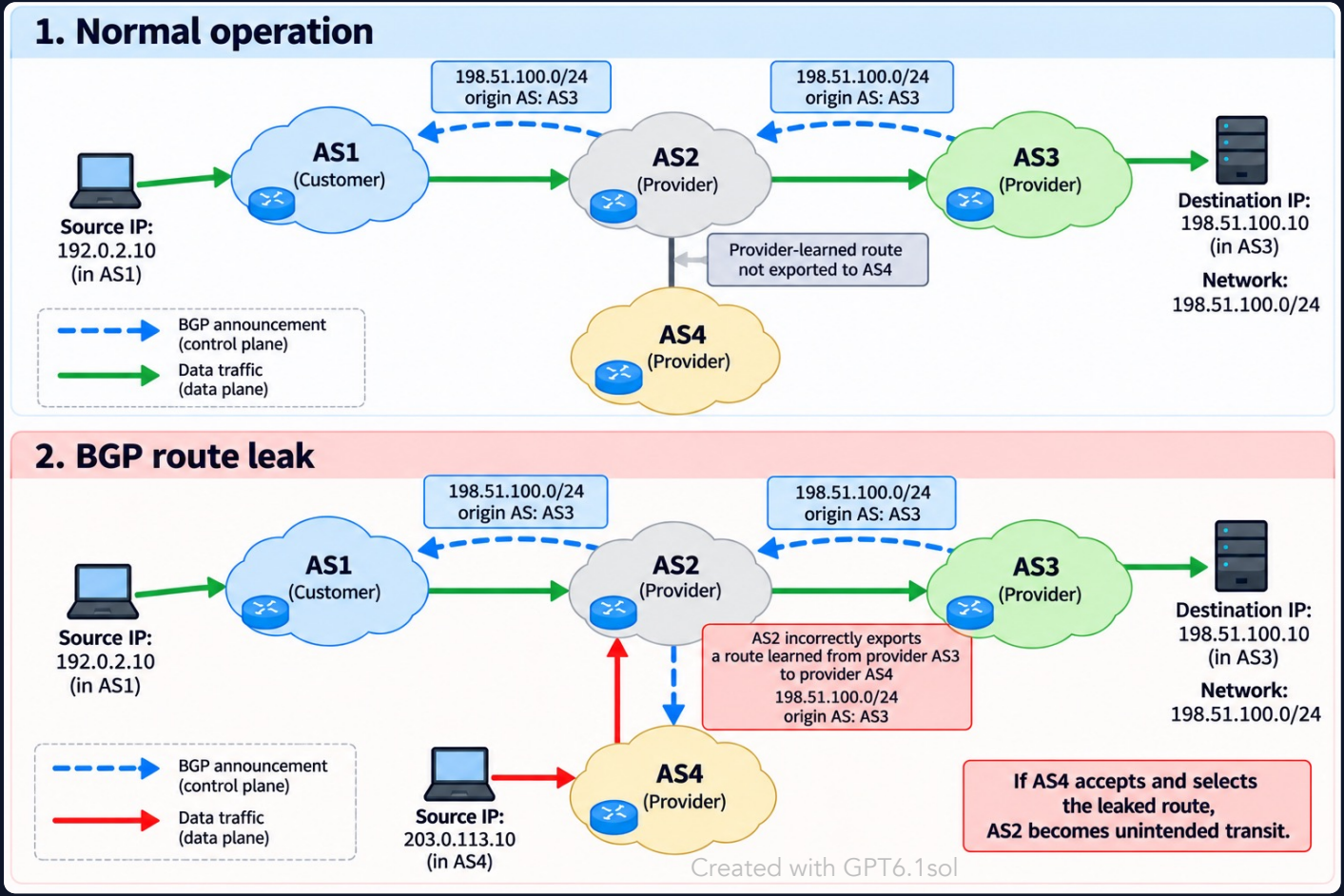
BGP Security Vulnerabilities
Example of Route Leaks



Normal operation (top diagram)

- AS3 legitimately originates a BGP announcement for 198.51.100.0/24.
- The announcement propagates from AS3 → AS2 → AS1 (blue dashed arrows).
- AS1 learns that the destination network is reachable through AS2 and AS3.
- When the user (192.0.2.10) sends packets to the server, traffic follows AS1 → AS2 → AS3 (green arrows).
- Importantly, AS2 does not advertise the route learned from AS3 to AS4, because both AS3 and AS4 are providers of AS2.

BGP Security Vulnerabilities
Example of Route Leaks



BGP route leak (bottom diagram)

- AS2 incorrectly advertises the route learned from provider AS3 to its other provider, AS4 (blue dashed arrow).
- The advertised prefix and origin remain legitimate: 198.51.100.0/24, originated by AS3.
- However, the announcement violates the expected BGP export policy: a customer should not normally advertise routes learned from one provider to another provider.
- This is the route leak. Unlike prefix hijacking, AS2 is not falsely claiming ownership of the destination network.

BGP Security Vulnerabilities

3. BGP Session Hijacking

- attacker hijacks a BGP session by compromising underlying TCP session
- attacker must know TCP ports, seq #s, and IP addresses of BGP peers*
- attacker may insert, modify, or delete routing updates
- by injecting forged TCP packets or exploiting weak session auth
  - e.g., inject malicious routing updates, set RST flag (BGP Session Reset Attack)

Mitigation

- TCP-AO (Authentication Option) – for integrity/authenticity of TCP connections
- restrict access to TCP port 179 using Access Control Lists (ACLs)
- use random TCP ports and seq #s to make hijacking harder (but doesn't solve)
- use IPsec to encrypt and authenticate BGP sessions

* BGP routers establish p2p connections using TCP port 179

BGP Security Vulnerabilities

BGP Configuration Errors

- human errors or misconfigurations cause routing issues
- may mimic unintended attack



Mitigation

- Automation and Templates – use standardized configurations and automated deployment tools to minimize human error
- Route Testing – validate routing configurations in a controlled environment before deployment
- Monitoring and Auditing – regularly audit configurations and monitor routes for anomalies

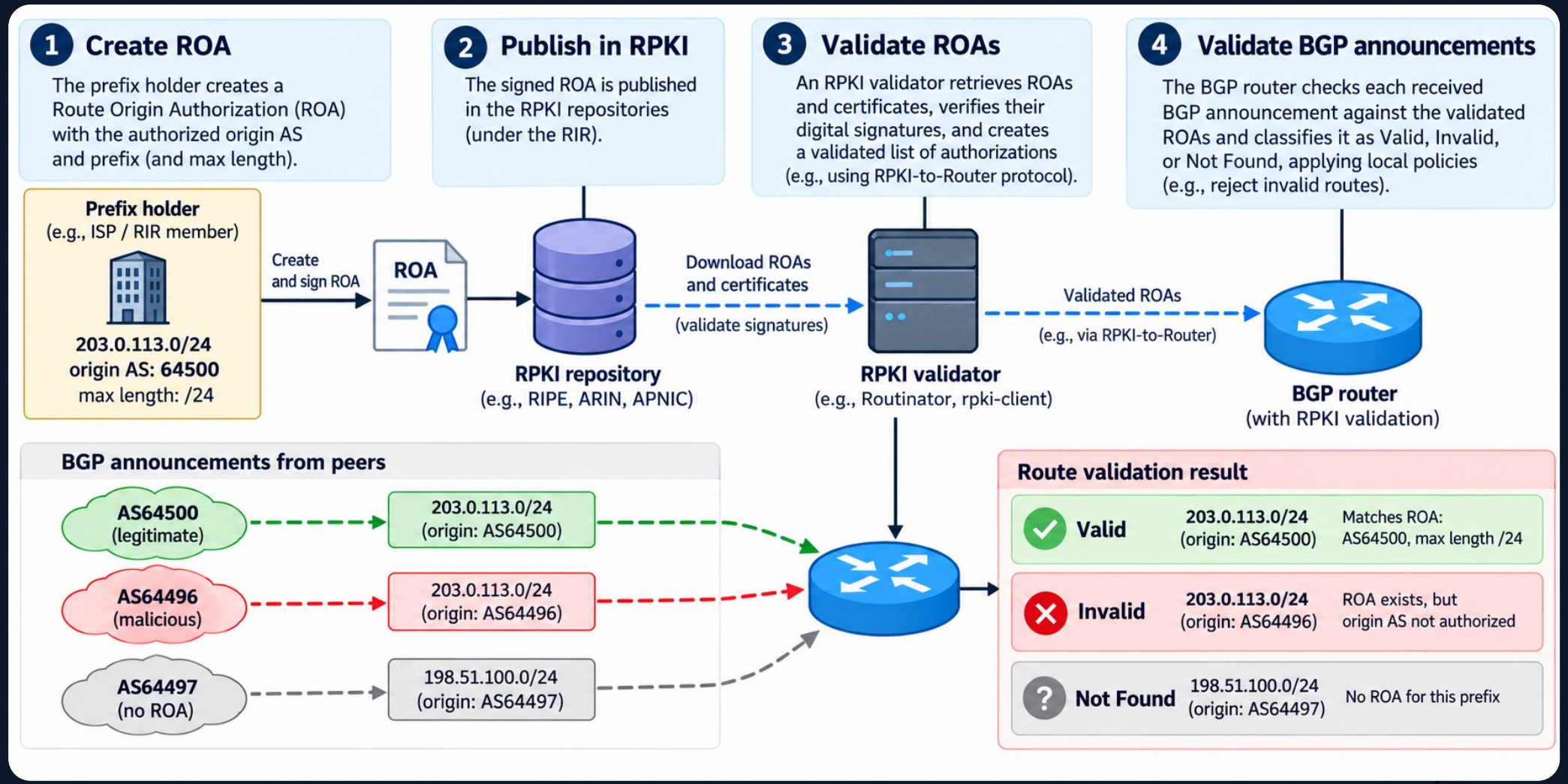
BGP Security Support Technologies

Resource Public Key Infrastructure (RPKI) [RFC 6811]

- specialized PKI framework to support improved security for BGP
- addresses the critical issue of route origin validation, mitigating the risk of prefix hijacking
- provides cryptographic proof of route origin validity using Route Origin Authorizations (ROAs)*
- BGP announcements are validated against a RPKI database through the interaction between routers and RPKI validators
- easy integration with the existing BGP-4 framework [RFC 4271]
- widely supported (e.g., Cisco, Huawei, Nokia) and adopted by ISPs
- requires robust management of the RPKI certificate hierarchy

* ROA object: (Origin AS, Network Prefix (e.g., 203.0.113.0/24), Max Prefix Length, Signature)

BGP Security Support Technologies
BGP RPKI Infrastructure Validation



[Cisco, adapted]

BGP Security Support Technologies

BGPsec (Extension to BGP-4) [RFC 8205]

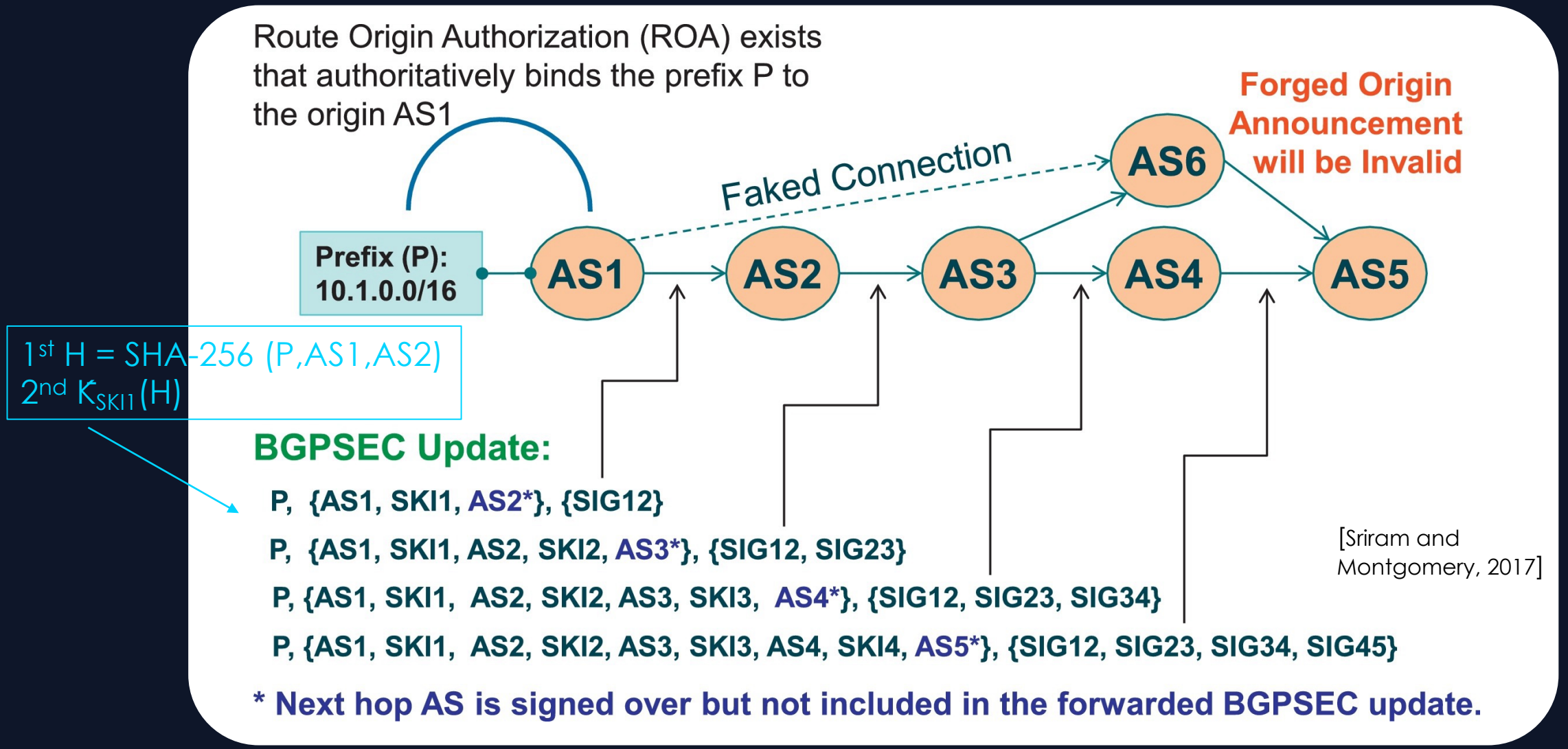
- addresses specific BGP vulnerability: lack of AS path validation
- enhances the authenticity and integrity of routing announcements by cryptographically signing BGP route updates
- validates if the AS path advertised in a BGP update reflects the actual path through which the prefix announcement traveled (chain of trust)
- prevents malicious actors from altering AS path attributes during transit
- complements RPKI by providing cryptographic AS path validation

  - ***while RPKI verifies the legitimacy of the origin AS for an IP prefix, BGPsec ensures the integrity of the AS path***

BGP Security Support Technologies: BGPsec

- Cryptographic AS Path Validation
  - each AS in the path signs the routing update using its private key (which is linked to its RPKI certificate through a Subject Key Index) before forwarding it
  - signature over: AS #, advertised prefix, next AS in the path and signature from prev AS
- Public Key Infrastructure (PKI)
  - uses PKI framework for signing and verifying updates
  - certificates are issued to ASes to prove their identities, tying AS #s to their public keys
- Backward Compatibility
  - routers not supporting BGPsec will simply propagate BGP updates w/o validation
- Incremental Deployment
  - allows for partial adoption, meaning only a subset of ASes need to implement it initially, and its benefits will gradually scale as adoption increases
- Increased Overhead
  - use of cryptographic signatures adds computational and bandwidth overhead

BGP Security Support Technologies



[Sriram and Montgomery, 2017]

BGP Security Support Technologies

Autonomous System Provider Authorization (ASPA) (Internet draft, Oct 25)

- addresses BGP route leaks, a vulnerability not fully mitigated by RPKI or BGPsec AS Path Validation
- introduces an additional validation layer for BGP updates without the complexity of AS path cryptographic signatures of BGPsec
- similarly to RPKI, ASPA verifies the legitimacy of the AS path based on a globally available database of authorized provider relationships
- BGP routers can validate if an AS path conforms to the expected routing model (P2C, C2P, P2P):
  - providers and peers do not transit traffic between other providers or peers, without explicit business agreement

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Image created with ChatGPT (OpenAI GPT-6.1)

Interior Gateway Protocols (IGPs)

Intradomain Routing

- routes the flow of packets within a single AS
- Interior Gateway Protocols (IGPs) ensure efficient, stable, and reliable routing within AS maintaining correct and up-to-date routing tables

- OSPF (Open Shortest Path First) [RFC 2328; RFC 5340]
  - link-state routing protocol widely used in enterprise and ISP networks
- IS-IS (Intermediate System to Intermediate System) (OSI) [RFC 9479]
  - link-state protocol, often deployed in ISP backbones
- EIGRP (Enhanced Interior Gateway Routing Protocol) [RFC 7868]
  - Cisco-proprietary hybrid routing protocol
- RIP (Routing Information Protocol) [RFC 2453]

Interior Gateway Protocols (IGPs): Intradomain Routing

Link-state vs. distance-vector

- Link-state (OSPF, IS-IS): Routers build a representation of the network topology and independently calculate the best paths.
- Distance-vector (RIP): Routers exchange routing information with neighbors, without maintaining a complete topology map.

Connection to routing security

- Just like BGP, IGPs rely on routing information received from other routers.
- If an attacker manipulates this information, they may redirect traffic or disrupt connectivity within the AS. [RFC 2453]

IGPs Security

Importance of Security in Intradomain Routing

- intradomain routing is trusted by design, assuming all routers belong to and are controlled by the AS operator
- insider threats, misconfigurations, or compromised routers can exploit this trust
- compromised IGP can disrupt internal traffic flow, affect the entire network or enable external attacks
- Examples:
  - malicious updates could blackhole traffic or redirect it for eavesdropping
  - instability caused by route flapping or false route advertisements (RAs)
- security challenges focus on the integrity of topology and routing updates

IGPs Vulnerabilities and Attacks

Lack of Confidentiality

- protocols like OSPF and IS-IS do not encrypt their updates by default, exposing sensitive topology information
- allows passive sniffing of routing packets on the network
- may result in:
  - leakage of internal network topology
  - helping attackers in planning further exploits

Mitigation

- encrypt routing traffic between routers using, e.g., IPsec ESP

IGPs Vulnerabilities and Attacks

Replay Attacks

- attacker may resend valid but old RAs*, tricking routers into using outdated or incorrect path
- attack involves capturing routing protocol packets (e.g., OSPF LSAs, IS-IS LSPs), delaying or replaying those packets to disrupt the network topology
- may result in traffic instability, looping, or blackholing, degrading perf.

Mitigation

- ensure each update includes a sequence number and a TTL value (e.g., `LSA Header` and `LSA Age` / `Max Age` in OSPF) so routers ignore packets with old sequence numbers or expired timers
- use cryptographic authentication: HMAC-SHA256 or other mechanisms to ensure packets are from trusted sources

*Router Advertisements

IGPs Vulnerabilities and Attacks

Spoofing and Injection Attacks

- attackers inject unauthorized, fake RAs to modify the network topology
- attack involves either gain access to a router or impersonate a legitimate router to inject false LSAs (OSPF)* or LSPs (IS-IS)* to advertise non-existent or malicious paths
- may result in traffic misdirection (e.g., redirecting traffic to attacker-controlled networks), creation of routing loops or blackholes

Mitigation

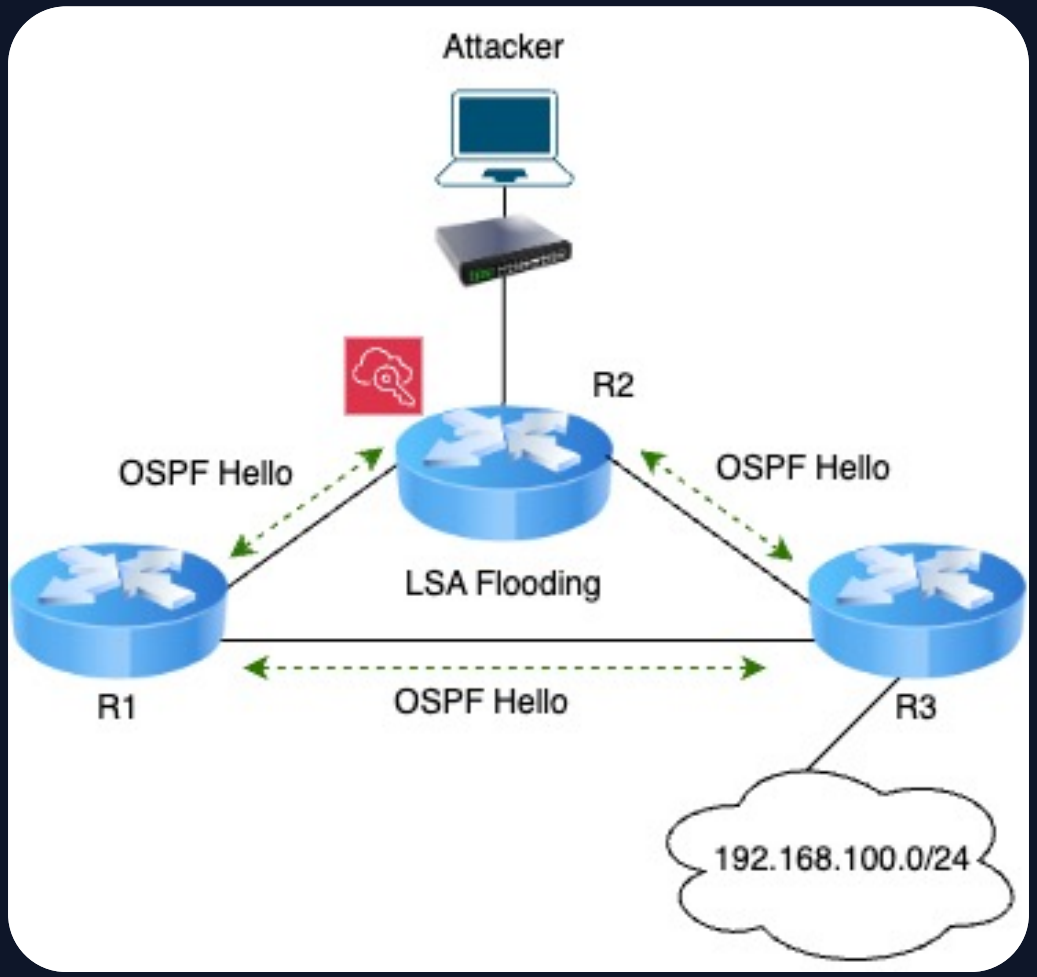
- authentication mechanisms – HMAC-SHA256 to authenticate LSAs (OSPF) or LSPs (IS-IS)
- secure management practices – restrict physical and logical access to routers
- encrypt and authenticate all routing comms using IPsec (adds complexity)

* LSA: Link-State Advertisement; LSP: Link-State Packet; Both LSA and LSP are the data packets or advertisements used by Link-State routing protocols to build a map of the network topology.

IGPs – Example of OSPF LSA Injection Attack

1. Pre-Attack: normal OSPF operation

- OSPF Hello Exchange
  - routers exchange Hello packets to establish and maintain neighbor relationships (R1 ↔ R2 ↔ R3)
- LSA Flooding
  - each router advertises its topology information using LSAs
  - e.g., R3 advertises net 192.168.100.0/24; R1 and R2 receive and store this info in their Link-State Databases (LSDBs)
- Route Calculation
  - routers calculate the shortest path to subnet using Dijkstra's algorithm, preferring R3 as the legitimate next hop



IGPs – Example of OSPF LSA Injection Attack

Step1: attacker captures LSAs

- either to (i) sniffing traffic or (ii) direct access to R2's LSDB if compromised
- attacker learns the topology

Step2: crafts malicious LSA

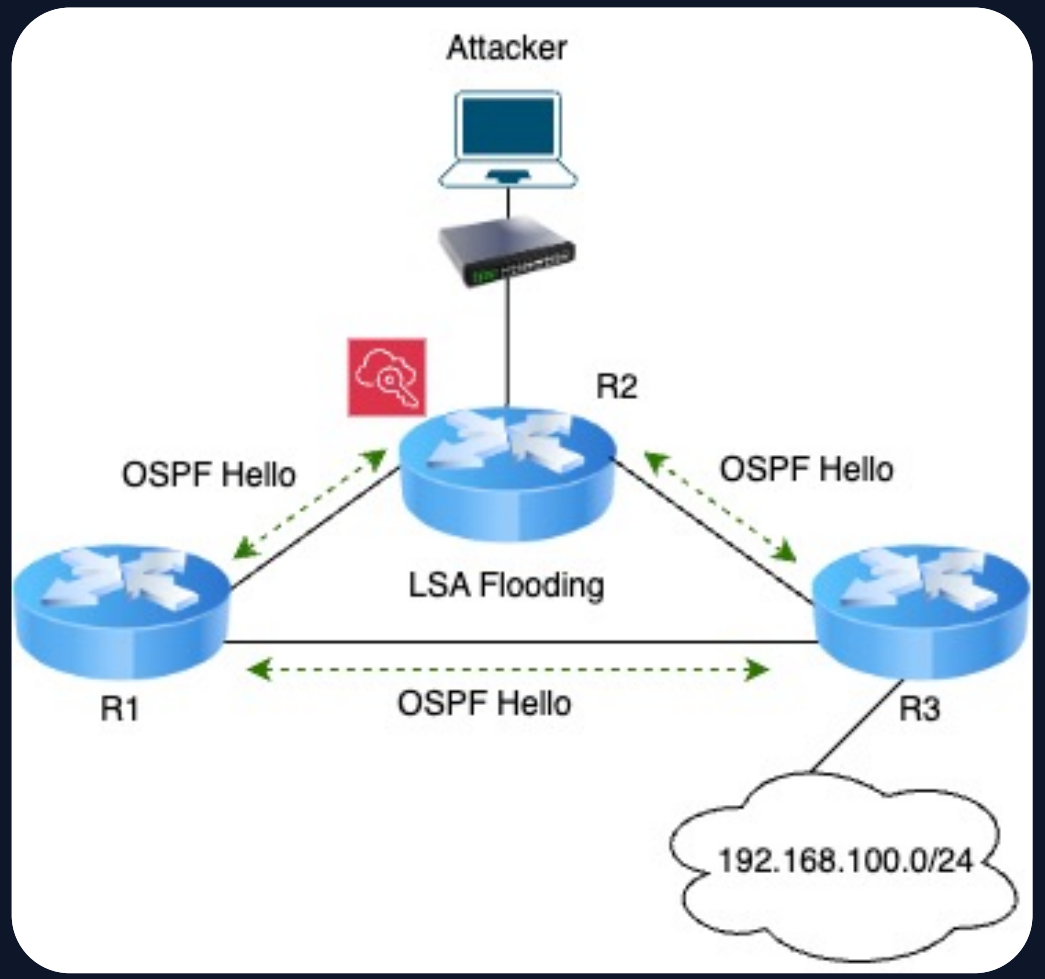
- sets lower metric for the subnet, making R2 appear to be the best route
- sets legitimate-looking sequence #s and checksum values to avoid rejection

Step3: injects malicious LSA

- attacker uses R2 to flood the malicious LSA to its neighbors (R1 and R3)

Step4: routing update; redirection

- R1 routes traffic for 192.168.100.0/24 through R2 instead of R3



Interior Gateway Protocols (IGPs): Summary

Protocol	Core RFC	Security RFCs	Principle	Notes
OSPFv2	RFC 2328	RFC 5709 (HMAC-SHA), RFC 7474 (IPsec)	Link-State	Widely used in IPv4 networks
OSPFv3	RFC 5340	RFC 7474 (IPsec for OSPFv3)	Link-State	Adds support for IPv6
IS-IS	RFC 9479	RFC 5304, RFC 5310	Link-State	Dual support for OSI and IP
EIGRP	RFC 7868	No specific RFC (MD5/HMAC [Cisco])	Hybrid (Link-State + Distance-Vector)	Cisco proprietary, less standardized
RIP	(RFC 1058), RFC 2453 (RIPv2)	No specific security RFC; supports MD5 in RIPv2	Distance-Vector	Simpler, less efficient; used for small networks

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Quizz

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